

HS Harz FuSa, Friedrichstraße 57

Airbag ECU

Messung und Verarbeitung der Sensoreingänge zur
Auslösung des Airbags
Measurement and Processing of Sensor Signals to Process
an Airbag Operation

Phase 8: Hardware Design

Version: 1.00

Contents

1	Meetings within the project.....	3
2	Unresolved topics.....	4
3	Glossar	5
4	Hardware Design	6
4.1	Hardware Architecture Block Description	6
4.2	Microcontroller Selection	7
4.3	Power Supply Design.....	7
4.4	CAN Communication Hardware	8
5	Figure and table listing.....	9
6	List of dependent documents	10
7	Literature	11
8	Version history	12

1 Meetings within the project

Topic	Time	Room	Participants
Item Definition	18:00	Zoom	ALL
Bericht:	<u>none</u>		

2 Unresolved topics

No unresolved topics

3 Glossar

Tabelle 1: Glossar

Begriff / Term	Erklärung / Explanation
ACU	Airbag Control Unit the ECU responsible for crash detection and airbag deployment control
ADC	Analog-to-Digital Converter converts analog sensor voltage to a digital integer value
ARM Cortex-M3	32-bit RISC processor core used in the Arduino Due (SAM3X8E)
ASIL	Automotive Safety Integrity Level A to D, D being highest requirement (ISO 26262)
CAN	Controller Area Network automotive serial communication bus standard (ISO 11898)
ECU	Electronic Control Unit embedded computing module in a vehicle
ESD	Electrostatic Discharge overvoltage event that can damage electronic components
EMI	Electromagnetic Interference unwanted electrical noise affecting circuit operation
GPIO	General Purpose Input/Output configurable digital pin on a microcontroller
HW	Hardware
ISO 26262	International standard for functional safety of road vehicle electrical/electronic systems
LDO	Low Drop-Out Regulator linear voltage regulator with low input-output voltage differential
MCU	Microcontroller Unit
MEMS	Micro-Electro-Mechanical System miniature sensor technology used for accelerometers and gyroscopes
MOSFET	Metal-Oxide-Semiconductor Field-Effect Transistor used as switching element in output drivers
Safing Condition	A secondary, independent confirmation of crash event required before deployment is enabled (dual-point safety architecture)
SWR	Software Requirement functional requirement assigned to the software module
SW	Software
TVS	Transient Voltage Suppressor protection diode for input overvoltage clamping

4 Hardware Design

This document covers the Hardware Design for the Airbag ECU prototype, documenting the physical sensor interfaces, microcontroller selection, power supply design, output driver stages, and CAN communication hardware.

4.1 Hardware Architecture Block Description

The hardware is organised into six primary functional blocks.

Tabelle 2 Hardware Architecture Block Description

Block	Component (Prototype)	Function
Power Supply	12 V to 5 V / 3.3 V LDO	Provides regulated voltage; capacitor hold-up for post-crash operation
Microcontroller	Arduino Due (ARM Cortex-M3)	Central processing: sensor reading, crash logic, output control, CAN
Sensor Inputs	Analog potentiometers (simulation)	Frontal acceleration (A0), gyroscope/rollover (A1), side pressure (A2)
Digital Input	Push button / switch (D2)	Baby/passenger seat detection signal (INPUT_PULLUP)
Output Drivers	LEDs on D8, D9, D13	Driver airbag, passenger airbag (simulated), warning lamp
CAN Interface	CAN0 (Due built-in) + MCP2551 transceiver	ISO 11898 CAN bus communication; diagnostic frame transmission and reception

Fig 01: Arduino Connections Figure 1 shows the complete Arduino prototype wiring: three analog potentiometers on A0/A1/A2 simulating the acceleration, gyroscope and pressure sensors, push button on D2 for the baby seat switch, and LEDs on D8/D9/D13 for the airbag and warning lamp outputs.

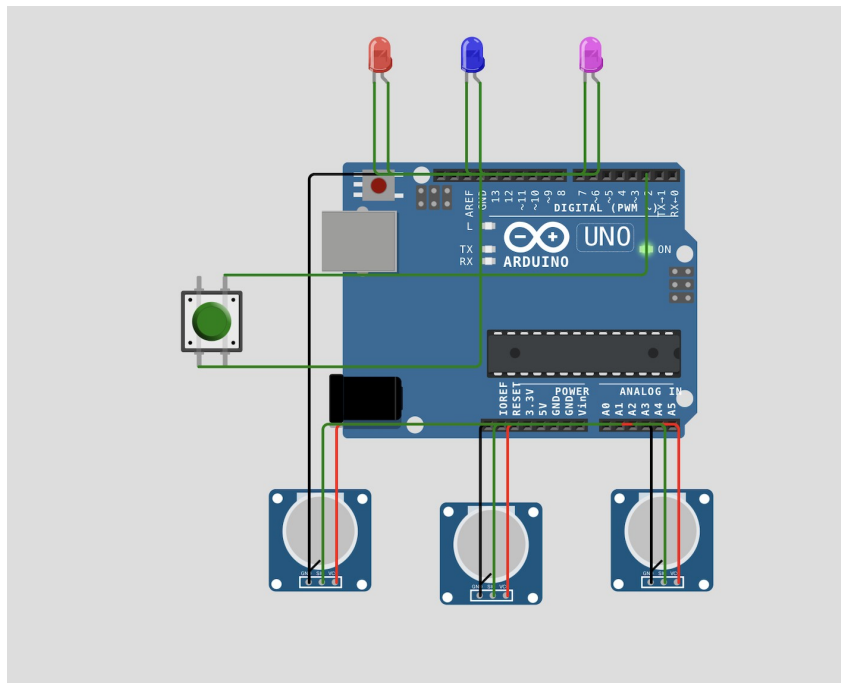


Abbildung 1 Arduino Connections

4.2 Microcontroller Selection

The Arduino Due is selected as the prototype microcontroller platform.

Tabelle 3 Arduino Due Specifications

Feature	Specification
CPU Core	ARM Cortex-M3, 32-bit
Clock Speed	84 MHz
Flash Memory	512 KB
SRAM	96 KB
Digital I/O Pins	54 (of which 12 PWM capable)
Analog Input Pins	12 (10-bit ADC, 0-3.3 V)
CAN Controllers	2x on-chip CAN 2.0B (CAN0, CAN1)
Operating Voltage	3.3 V I/O logic (5 V tolerant inputs)
USB Interface	Native USB + Programming USB

In a production-grade Airbag ECU, a qualified automotive-grade microcontroller (e.g., Infineon AURIX TC3xx, NXP S32K, or Renesas RH850) with hardware safety mechanisms including lock-step CPU cores, ECC memory, hardware watchdog, and SIL/ASIL-certified software toolchains would be selected. The Arduino Due is used exclusively as a functional prototype. 4.3 Sensor Pinout and Thresholds

Tabelle 4 Sensor Inputs and Thresholds

Sensor	Pin	ADC Range	Crash Threshold	Function
Acceleration Sensor (frontal crash)	A0	0-1023 (10-bit ADC)	≥ 700	Triggers frontal crash condition
Gyroscope Sensor (rollover / safing)	A1	0-1023 (10-bit ADC)	≥ 600	Required as safing channel (dual-point logic)
Pressure Sensor (side impact)	A2	0-1023 (10-bit ADC)	≥ 650	Triggers side crash condition
Baby/Passenger Seat Switch	D2	Digital Active LOW (INPUT_PULLUP)	LOW = baby seat	Suppresses passenger airbag deployment

Tabelle 5 Baby/Passenger Seat Switch Parameters

Parameter	Detail
Arduino Pin	D2 (Digital Input)
Logic Type	Active LOW with internal INPUT_PULLUP enabled
LOW (0 V)	Baby/child seat detected passenger airbag suppressed
HIGH (3.3 V)	No baby seat detected passenger airbag deployment permitted
Fault Condition	Signal outside valid range or intermediate voltage safe state

4.3 Power Supply Design

The Airbag ECU power supply must remain functional during and after a crash event.

Tabelle 6 Power Supply Architecture

Supply Stage	Specification
Input Voltage Range	9-16 V DC (vehicle battery nominal 12 V)
Primary Regulator	5 V LDO or switching DC-DC converter for MCU supply
MCU Core Supply	3.3 V (Arduino Due operates at 3.3 V logic)
Post-Crash Hold-Up	Capacitor bank (≥ 150 ms at full load) powers squib firing and MCU after battery loss
Reverse Polarity Protection	P-channel MOSFET or Schottky diode at battery input
Overvoltage Protection	TVS diode at input to clamp transients (load dump: up to 40 V per ISO 7637-2)
Prototype Supply	Arduino Due powered via USB (5 V) with onboard 3.3 V LDO; no crash hold-up in prototype

4.4 CAN Communication Hardware

Tabelle 7 CAN Bus Interface

CAN Parameter	Value
CAN Standard	ISO 11898 / CAN 2.0B
CAN Controller	On-chip CAN0 (Arduino Due SAM3X8E)
CAN Transceiver (Production)	MCP2551 or TJA1050 (5 V, ISO 11898 compliant)
Message ID	0x120 (high priority airbag frame)
Data Length Code (DLC)	8 bytes
Typical Bus Speed	500 kbit/s (automotive standard)
Termination	120 Ohm at each end of the CAN bus

Tabelle 8 CAN Frame Structure

Byte	Field Name	Content / Meaning
Byte 0	System Status	0x00 = Normal, 0x01 = Crash detected, 0xFF = Fault
Byte 1	Acceleration Value	Raw ADC reading from A0 (0-255 scaled)
Byte 2	Gyroscope Value	Raw ADC reading from A1 (0-255 scaled)
Byte 3	Pressure Value	Raw ADC reading from A2 (0-255 scaled)
Byte 4	Seat Status	0x00 = No baby seat, 0x01 = Baby seat detected
Byte 5	Deployment Status	Bit 0 = Driver airbag, Bit 1 = Passenger airbag
Byte 6	Fault Code	0x00 = No fault, 0x01 to 0xFF = Specific fault codes
Byte 7	Checksum	XOR checksum of Bytes 0-6 for frame integrity

5 Figure and table listing

Tabelle 1: Glossar	5
Tabelle 2 Hardware Architecture Block Description.....	6
Tabelle 3 Arduino Due Specifications.....	7
Tabelle 4 Sensor Inputs and Thresholds	7
Tabelle 5 Baby/Passenger Seat Switch Parameters	7
Tabelle 6 Power Supply Architecture.....	8
Tabelle 7 CAN Bus Interface	8
Tabelle 8 CAN Frame Structure	8
 Abbildung 1 Arduino Connections	 6

6 List of dependent documents

No.	Dokument name	Version	Date	File name
1	Airbag ECU Hardware Concept Design Report	V02	22.05.2026	AirbagECU_HW_Concept_Design_Report_V02.pdf
2	ECU Airbag System Architecture Documentation	V02	27.05.2026	ECU Airbag system architecture documentation V2.pdf
3	Arduino Coding and Logic Implementation for Airbag ECU System	V1	02.06.2026	FuSa Report_Adruido.docx

7 Literature

No. Reference

- [1] International Organization for Standardization (ISO), ISO 26262: Functional Safety Standard for Modern Road Vehicles. https://fscdn.rohm.com/en/products/databook/white_paper/iso26262_wpe.pdf
- [2] S. N. Easwaran, M. Mollat, D. Sreedharan et al., Fault Tolerant Low Side Driver Circuit Design Using Design FMEA for Automotive Applications, IEEE NEWCAS, 2019. doi: 10.1109/newcas44328.2019.8961225
- [3] G. Bahig and A. El-Kadi, Formal Verification of Automotive Design in Compliance With ISO 26262, IEEE Access, vol. 5, pp. 4505–4516, 2017. doi: 10.1109/access.2017.2683508
- [4] M. R. Kabir, S. Boddupalli, A. P. D. Nath, and Sandip Ray, Automotive Functional Safety: Scope, Standards, and Perspectives on Practice, IEEE Consumer Electronics Magazine, vol. 14, no. 1, pp. 10–25, 2025. doi: 10.1109/mce.2024.3394165
- [5] S. Noh, Enhancing AUTOSAR safety mechanisms for ISO 26262 functional safety requirements, International Journal of Automotive Technology, vol. 17, no. 4, pp. 715–723, 2016.
- [6] M. U. Ali et al., Intelligent Driver Drowsiness Detection for Road Safety.
- [7] U. Jain, Software-in-the-Loop (SiL) Simulation and Validation of Airbag ECU, I4Tech, 2025. doi: 10.1109/i4tech64670.2025.11277665
- [8] R. Mariani, RISC-V Functional Safety for Autonomous Automotive Systems, arXiv:2604.17391, 2026. doi: 10.48550/arXiv.2604.17391
- [9] P. Kilian, O. Koller, P. Van Bergen et al., Emergency Operation in the Power Supply Domain According to ISO 26262, IEEE Access, vol. 10, pp. 47557–47569, 2022. doi: 10.1109/ACCESS.2022.3170903
- [10] Lee, K.J., Ki, Y.H., Cheon, J.S. et al., Approach to functional safety-compliant ECU design for electro-mechanical brake systems, Int. J. Automot. Technol., vol. 15, pp. 325–332, 2014. doi: 10.1007/s12239-014-0033-7
- [12] Y. Li, Q. Liu, and C. Cao, Integrated Safety Validation of Connected Automated Vehicles Using Distributed Hardware Redundancy Architectures, Security and Safety, vol. 2, p. 2022009, 2023. doi: 10.1051/sands/2022009

8 Version history

Document name	date	Change log
Airbag_ECU_Phase_4	14.06.2026	v1.00 Initial version. Hardware Design compiled from HW Concept Design Report V02 Sections 3, 4, 5, 6, 7, 8.